Gracie’s clock. He comes to the surprising conclusion that Gracie will receive his signals proclaiming the passing of an hour on his clock after the appointed hour has passed on hers. In fact, since George is aware of Gracie’s expertise in physics, he knows that she will take the signal’s travel time into account when drawing conclusions about his clock based on his cell phone communications. A little more calculation quantitatively shows that even taking the travel time into account, Gracie’s analysis of his signals will lead her to the conclusion that George’s clock ticks more slowly than hers.

Exactly the same reasoning applies when we take Gracie’s perspective, with her sending out hourly signals to George. At first the slowness of George’s clock from her perspective leads her to think that he will receive her hourly messages prior to broadcasting his own. But when she takes into account the ever longer distances her signal must travel to catch George as he recedes into the darkness, she realizes that George will actually receive them after sending out his own. Once again, she realizes that even if George takes the travel time into account, he will conclude from Gracie’s cell phone communications that her clock is running slower than his.

So long as neither George nor Gracie accelerates, their perspectives are on precisely equal footing. Even though it seems paradoxical, in this way they both realize that it is perfectly consistent for each to think the other’s clock is running slow.

Motion’s Effect on Space

The preceding discussion reveals that observers see moving clocks ticking more slowly than their own-that is, time is affected by motion. It is a short step to see that motion has an equally dramatic effect on space. Let’s return to Slim and Jim on the drag strip. While in the showroom, as we mentioned, Slim had carefully measured the length of his new car with a tape measure. As Slim is speeding along the drag strip, Jim cannot apply this method to measure the length of the car, so he must proceed in an indirect manner. One such approach, as we indicated earlier, is this: Jim starts his stopwatch just when the front bumper of the car reaches him and stops it just as the rear bumper passes. By multiplying the elapsed time by the speed of the car, Jim can determine the car’s length.